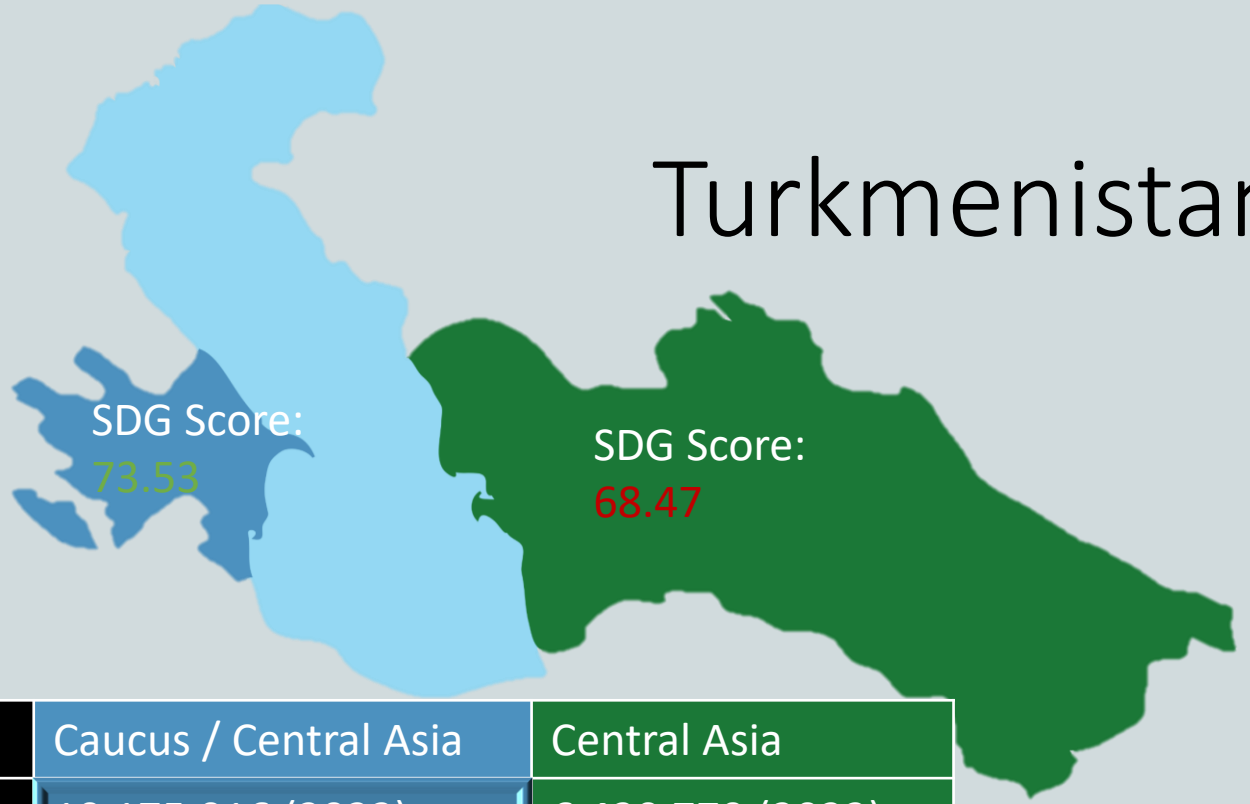
Environmental Analysis of Azerbaijan

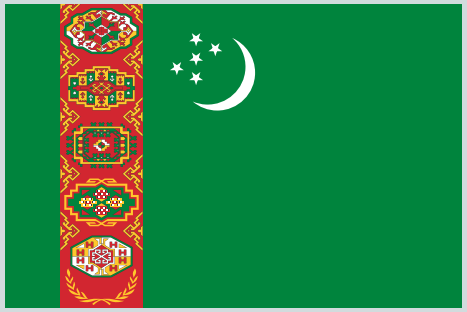
By: Peter Polner

Azerbaijan





Turkmenistan



Region	Caucus / Central Asia	Central Asia
Population	10,175,016 (2022)	6,430,770 (2022)
Population growth (%)	0.4% (2022)	1.4% (2022)
Population Density (sq. km)	123 (2021)	13 (2021)
GDP (billion USD)	78.72 (2022)	45.61 (2022)
GDP per Capita (USD)	7,736.7 (2022)	7,297.2 (2022)
Land Area (sq. km)	86370.73	490263.93

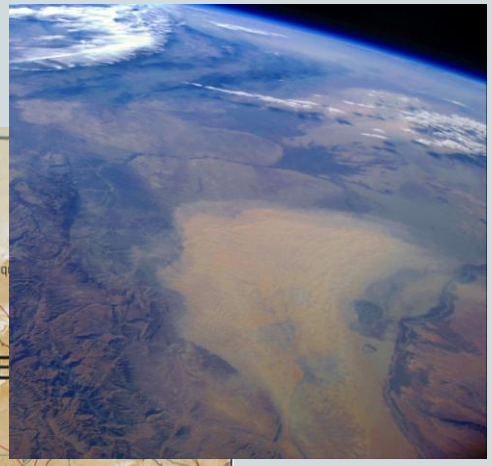
Data – World Bank Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://data.worldbank.org/?locations=AZ-TM
Environmental Performance index. Retrieved 11/13/2023 from https://epi.yale.edu/epi-results/2022/country/tkm, https://epi.yale.edu/epi-results/2022/country/aze
Sustainable Development report. Retrieved 11/13/2023 from https://dashboards.sdgindex.org/rankings

Geography & Climate





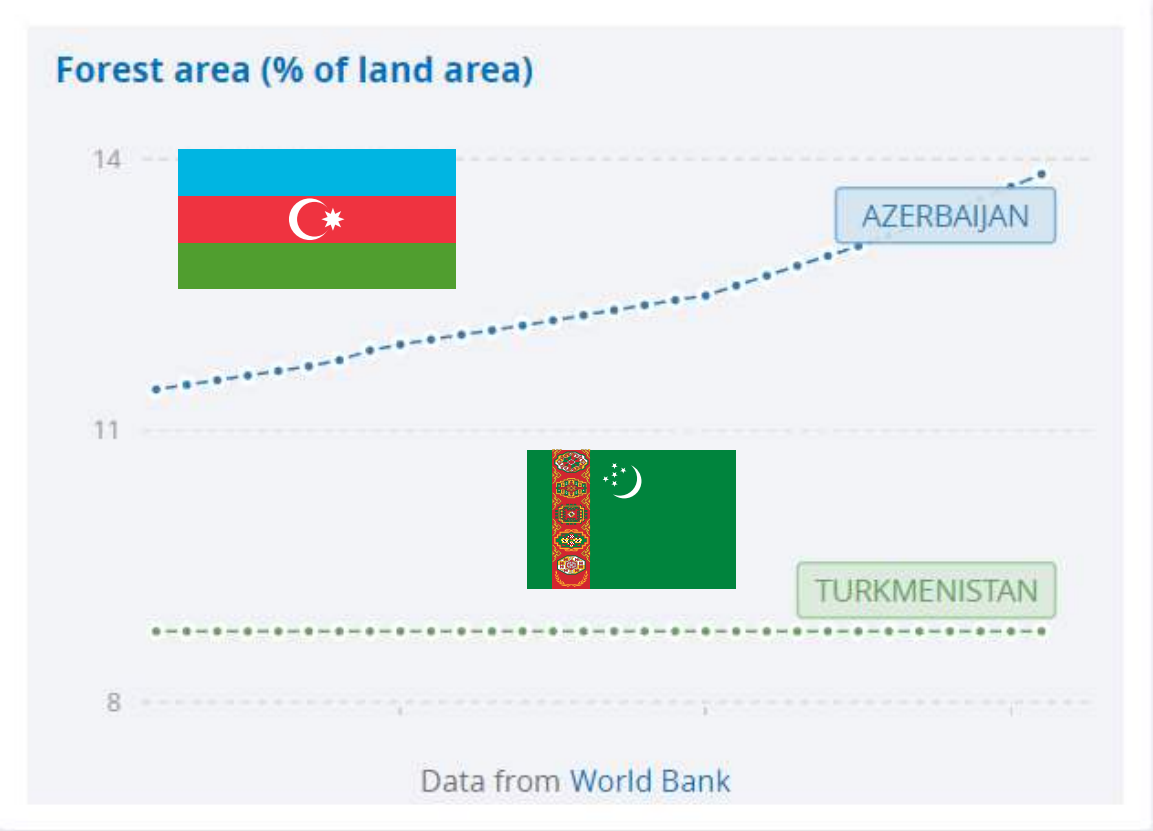




Azerbaijan	Turkmenistan
dry, semiarid steppe • Variable Climate • Surrounded by mountainous steppes • Access to Caspian Sea	subtropical desert • Mostly Inhospitable • Large desert covers country • Access to Caspian Sea

CIA World Factbook. Retrieved 11/13/2023 from https://www.cia.gov/the-world-factbook/field/climate

Forested land (Trend)



Azerbaijan

- 13.9% of land area
  - Less overall forests
- Clear Increasing Trend

Turkmenistan

- 9% of land area
  - More forests area overall
- No Growth
- Able to maintain stable forests (No decrease)

Data – World Bank Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://data.worldbank.org/indicator/AG.LND.FRST.ZS?locations=AZ-TM
Data – World Bank Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://data.worldbank.org/indicator/AG.LND.FRST.K2?locations=AZ-TM

Biodiversity – Azerbaijan



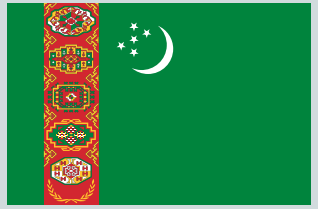
Factors	Actions
• Crossroad between Europe and Asia • Changing climate, land masses, & sea connections created isolation • Resulted in a unique species composition	• System of strict nature reserves • Parks opened for research & visitors • Corruption and other factors hinder further protection





USAID: Biodiversity analysis. Retrieved 11/13/2023 from https://www.researchgate.net/publication/343390648_The_Biodiversity_Analysis_for_Azerbaijan_2020

Biodiversity – Turkmenistan



Factors	Actions
• Fragile desert ecosystem • Supported by mountainous forest regions • Geological Caspian sea basin created isolation -> Diversity	• Creation of well-spread park system • Infrastructure removed from protected areas • Prevention of desertification by urban forestation unsuccessful (and hurt biodiversity) • Important role of biodiversity in research





USAID Biodiversity Assessment. Retrieved 11/13/2023 from https://pdf.usaid.gov/pdf_docs/PNACN473.pdf

Biodiversity – World Ranking

COUNTRY	RANK	EPI SCORE	10-YEAR CHANGE
Azerbaijan 	118	46.20	0.70
Turkmenistan 	156	22.90	0.60

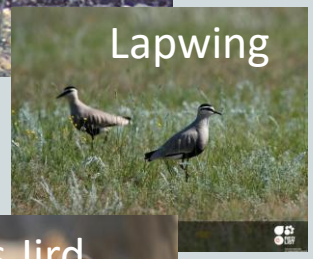
Ranks Biome Protection

Azerbaijan	Turkmenistan
• Better retention and protection of biodiversity • Improving more than Turkmenistan	• Comparatively way worse protection of biodiversity (over 30 ranks under Azerbaijan) • Positive 10y change

Environmental Performance Index. Retrieved 11/13/2023 from https://epi.yale.edu/epi-results/2022/component/bdh

Endangered Species

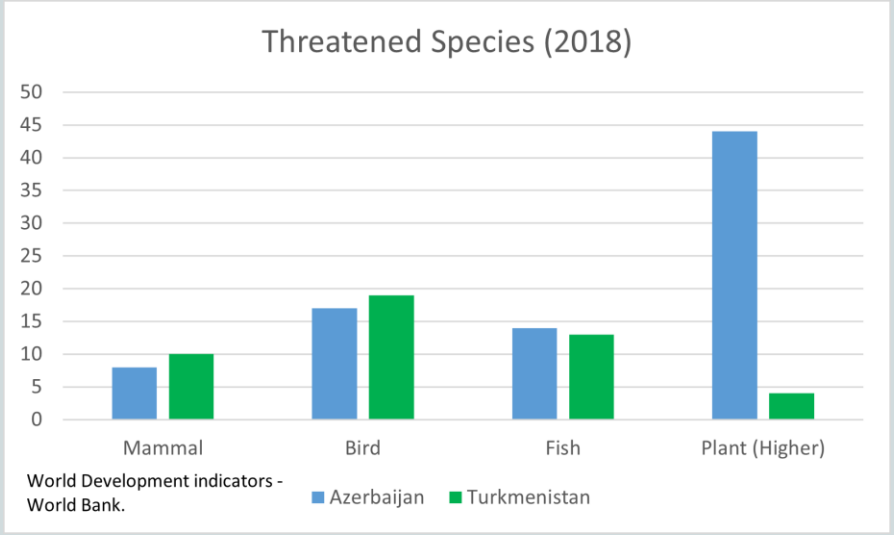






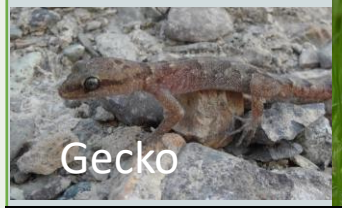


Azerbaijan	Turkmenistan
1. Slender Billed Curlew	1. Saiga
2. Sociable Lapwing	2. Siberian Crane
3. Dahl's Jird	3. Persian sturgeon
4. Azerbaijani Spring Roach	4. Southern Even-fingered Gecko
5. Beluga	5. Beluga









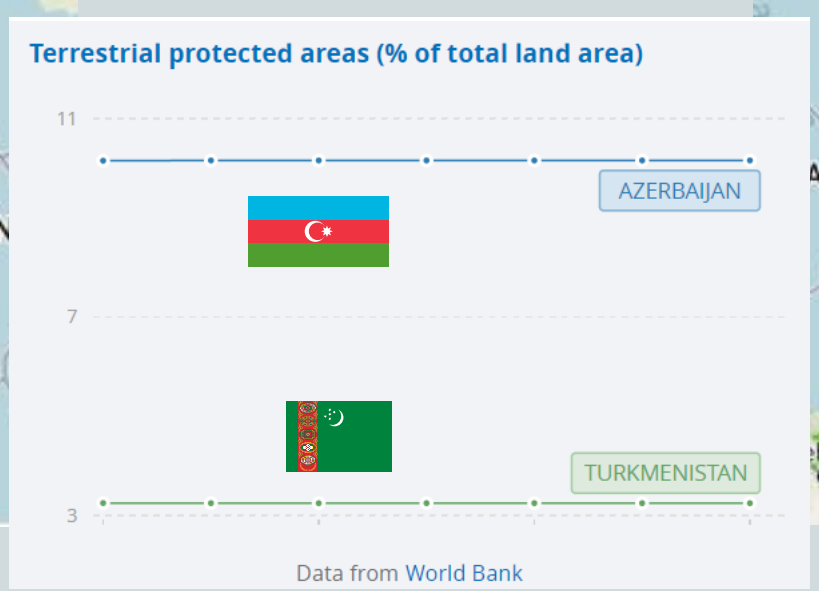




IUCN Red List. Retrieved 11/13/2023 from https://www.iucnredlist.org/search/grid?landRegions=AZ&searchType=species
IUCN Red List. Retrieved 11/13/2023 from https://www.iucnredlist.org/search?landRegions=TM&searchType=species
World Development indicators - World Bank. Retrieved 11/13/2023 from http://wdi.worldbank.org/table/3.4

Protected Areas





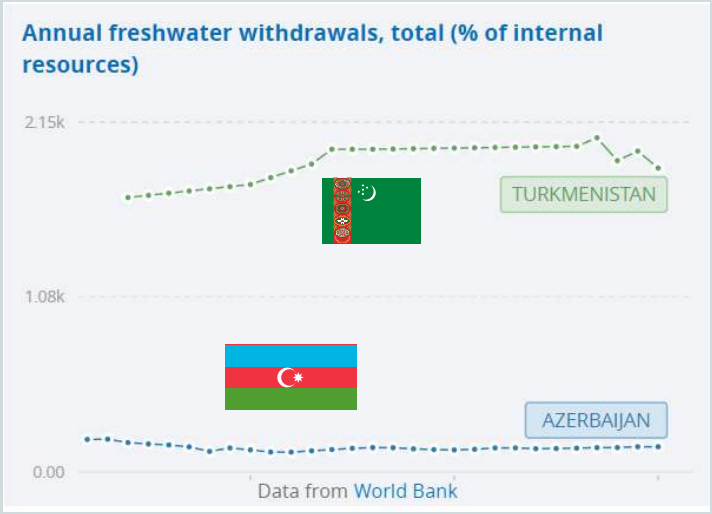


Azerbaijan	Turkmenistan
• 37 Protected Areas • 10% of land • Constant Trend	• 32 protected Areas • 3% of land • Constant Trend

Data – World Bank Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://data.worldbank.org/indicator/ER.LND.PTLD.ZS?end=2022&locations=AZ-TM&start=2016&view=chart
UNEP-WCMC Protected Area Profile for Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://www.protectedplanet.net/country/TKM & https://www.protectedplanet.net/country/AZE

Water







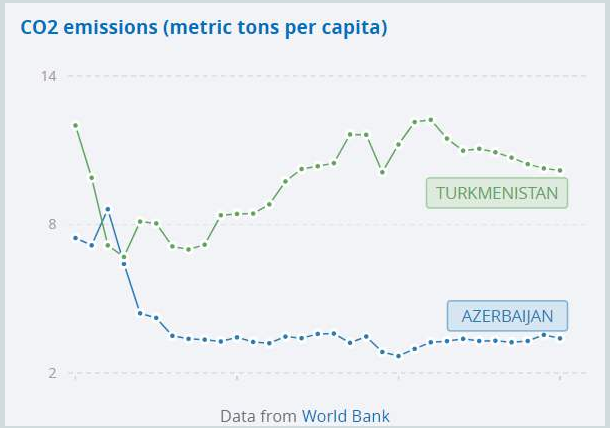
Country	Water withdrawals	Sanitation (Score)	Drinking Water (score)	fisheries
Azerbaijan	155	44.3	46.5	N/A
Turkmenistan	1868	46.9	47.7	N/A

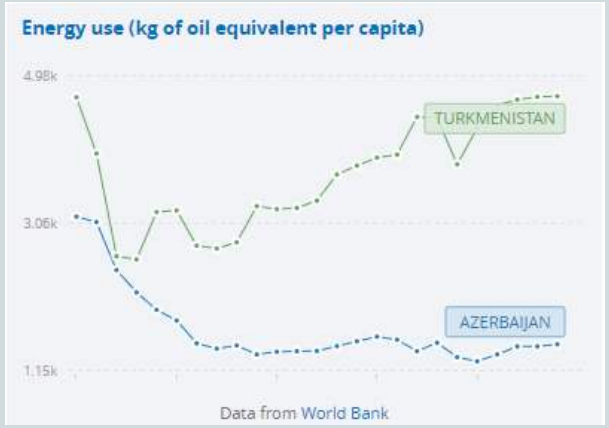
Azerbaijan	Turkmenistan
• Less water withdrawals (for more people) • Worse sanitation and drinking water • Fishing Data not reported	• Better Sanitation and drinking water (lower pop.) • Over 10x more water withdrawals than Azerbaijan • Fishing Data not reported

Data – WorldBank. Retrieved 11/13/2023 from https://data.worldbank.org/indicator/ER.H2O.FWTL.ZS?locations=AZ-TM
Environmental Performance Index: Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://epi.yale.edu/epi-results/2022/country/aze & https://epi.yale.edu/epi-results/2022/country/tkm

Energy & Co2 Emissions

Country	Energy Use (kg oil per cap.)	Energy Savings	Greenhouse Gasses (Rank)	Renewable Energy Use (%)	Climate & Energy EPI Rank
Azerbaijan	1502	NA	105	1.23	104
Turkmenistan	4723	NA	129	0.06	118





Azerbaijan

- Lower energy use
- Higher renewable profile
  - Less greenhouse gasses

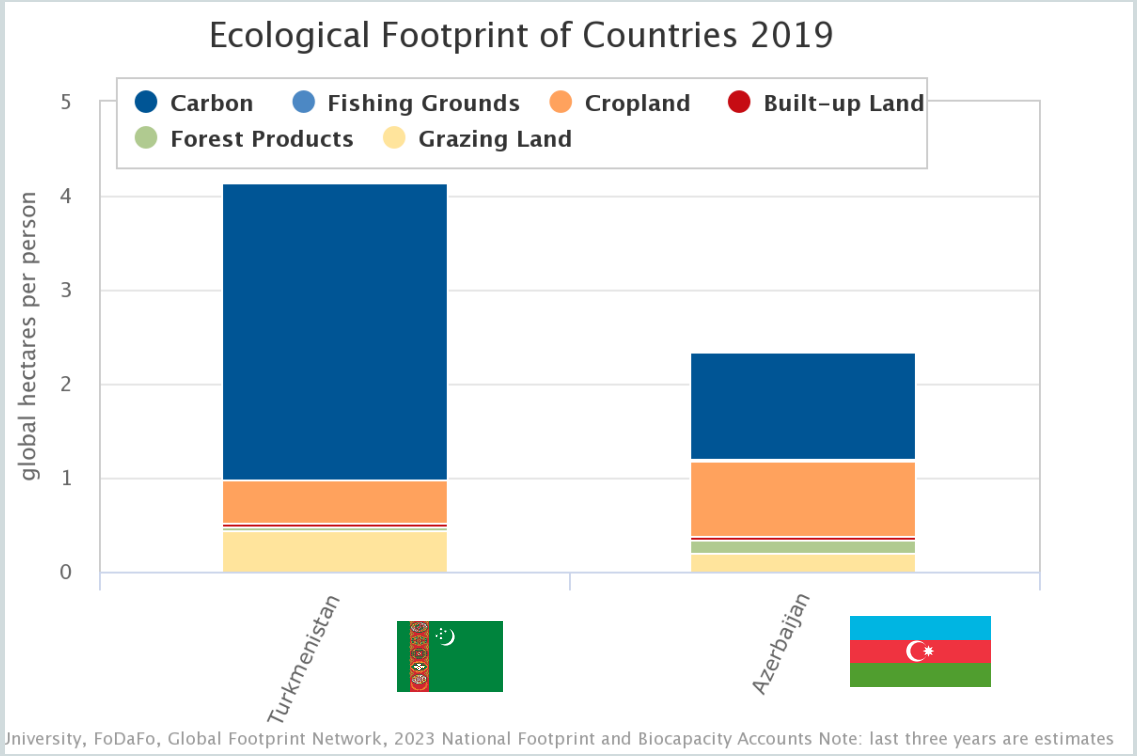
Turkmenistan

- High energy use
- Calming trend of GHGE
- Low renewable supply

Data – World Bank. Retrieved 11/13/2023 from https://data.worldbank.org/?locations=AZ-TM
Environmental Performance Index: Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://epi.yale.edu/epi-results/2022/country/aze & https://epi.yale.edu/epi-results/2022/country/tkm

Ranking

Azerbaijan	Turkmenistan
• Better environmental rank • Negative trend • Smaller ecological footprint	• Positive Trend • Larger carbon footprint



COUNTRY	RANK	EPI SCORE	10-YEAR CHANGE
Azerbaijan	104	38.60	-1.30
Turkmenistan	118	37.00	8.90

Environmental Performance Index: Azerbaijan & Turkmenistan. Retrieved 11/13/2023 from https://epi.yale.edu/epi-results/2022/country/aze & https://epi.yale.edu/epi-results/2022/country/tkm

Footprint Network Country Comparison. Retrieved 11/13/2023 from https://data.footprintnetwork.org/?_ga=2.263195461.836682970.1700466701-175830558.1700466701#/compareCountries?type=EFCpc&cn=52,213&yr=2019

SWOT

Strength

- Variable & Livable Climate
- Surrounded by mountainous steppes with access to Caspian Sea
- Large growing forests
- System of strict nature reserves
  - Parks opened for research & visitors
- Good and improving retention/protection of biodiversity
- Less threatened mammal and bird species
- Large amounts of protected areas at constant trend
- Low amounts of water withdrawals
- Low Energy Use
- High Renewables profile -> Low GHGE

Weakness

- Less land area
- Biodiversity protection hindered by corruption and other factors
- Marine and plant life heavily threatened
- bad sanitation and drinking water
- Fishing Data not reported

Azerbaijan 

Threats

- Desertification effects
- Corruption can create worse effects
- Land issues over growing cities (reduction of protected areas)
- Efforts to improve conditions (water), encroach on protected areas

Opportunities

- Growing forest areas combat desertification (non-urban)
- Growing market for renewable energy
- Biodiversity protection sought after

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- Artaev, Oleg & Levin, Boris & Mustafayev, Namiq & Simonov, Evgeniy. (2018). Caucasian Relict Fish - Azerbaijani Spring Roach. 2018.